Task-9:- Technical Documentation(Deploying Applications on AWS: VM-Based and Serverless Architectures)

1. Introduction

1.1 Purpose

The purpose of this technical document is to describe the design, implementation, deployment, and operational strategy of the **PowerOfMath Application** deployed on AWS using:

- Serverless Architecture (AWS Lambda + API Gateway + DynamoDB)
- Traditional VM-Based Architecture (EC2 + Flask + Nginx + DynamoDB)

This document is intended for:

- CTO
- Developers
- Solution Architects
- DevOps Engineers
- Resellers
- Cloud Architects

It provides complete visibility into infrastructure, DevOps practices, cost analysis, security implementation, and operational processes.

1.2 Scope

This document covers:

- AWS Infrastructure setup
- Application deployment using two architectures
- IAM configuration
- SSL setup
- Reverse proxy configuration
- Monitoring
- Security implementation
- Cost optimization
- DevOps best practices alignment with AWS Well-Architected Framework

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2. Overview of DevOps Project

2.1 Definition

DevOps is a collaborative approach combining development and operations to enable faster, reliable, and secure software delivery.

In this project:

- Code is version controlled
- Infrastructure is provisioned in AWS
- Deployment is automated
- Security and scalability are integrated

Benefits achieved:

- Faster deployment
- Reduced downtime
- Scalability
- Improved operational efficiency

2.2 Key Principles – AWS Well-Architected Framework

This solution follows the 6 pillars:

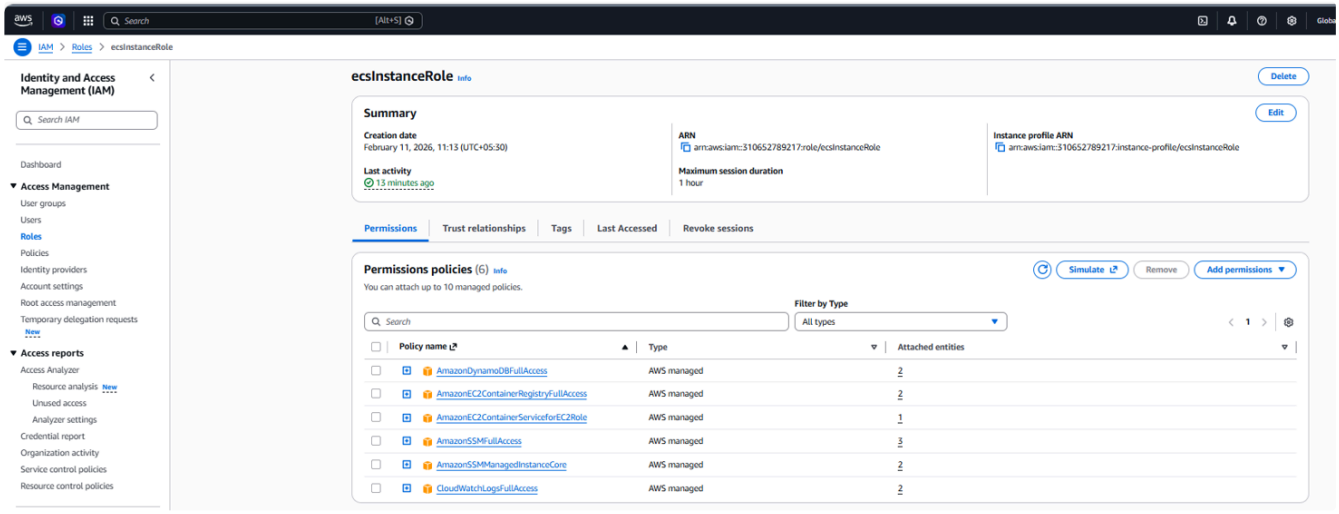
1️⃣ Operational Excellence

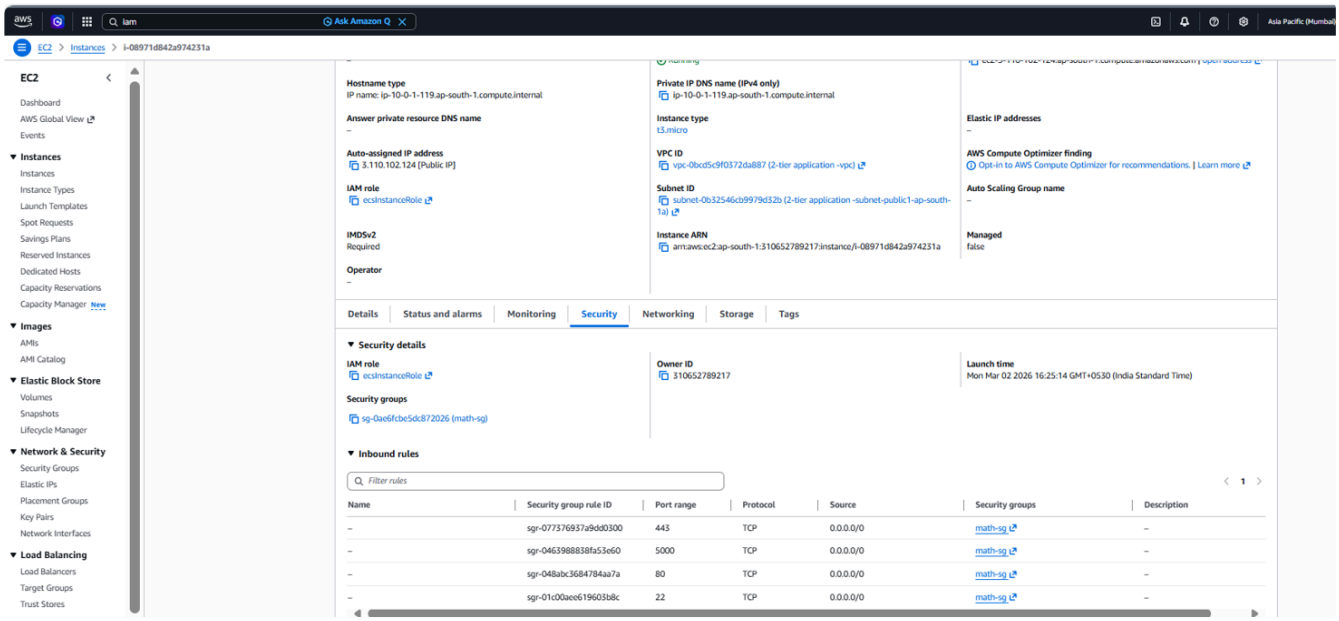
- Systemd used for service reliability
- Monitoring via logs
- Proper service restart policies

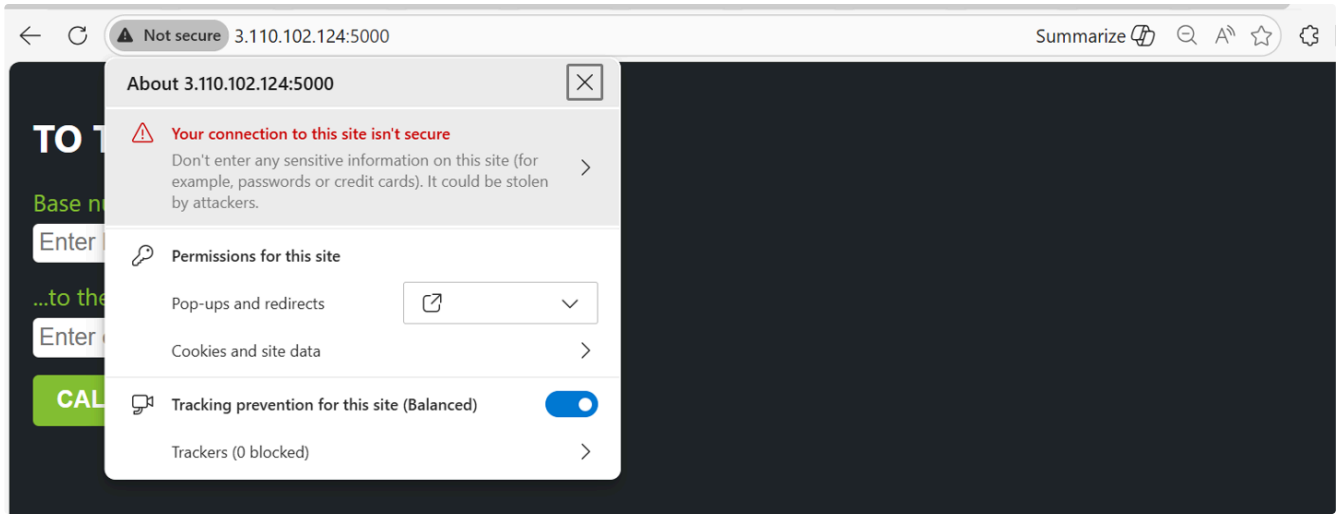
```
(venv) ubuntu@ip-10-0-1-119:~$ sudo systemctl start powerofmath
sudo systemctl status powerofmath
● powerofmath.service - PowerOfMath Flask App
     Loaded: loaded (/etc/systemd/system/powerofmath.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-02 12:59:09 UTC; 31ms ago
   Main PID: 12763 (python)
      Tasks: 1 (limit: 1008)
     Memory: 4.1M (peak: 4.1M)
        CPU: 26ms
     CGroup: /system.slice/powerofmath.service
             └─12763 /home/ubuntu/venv/bin/python /home/ubuntu/app.py

Mar 02 12:59:09 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
```

2️⃣ Security

- IAM roles attached to Lambda and EC2
- Security Groups configured
- HTTPS enabled via ACM & Certbot
- Principle of Least Privilege applied

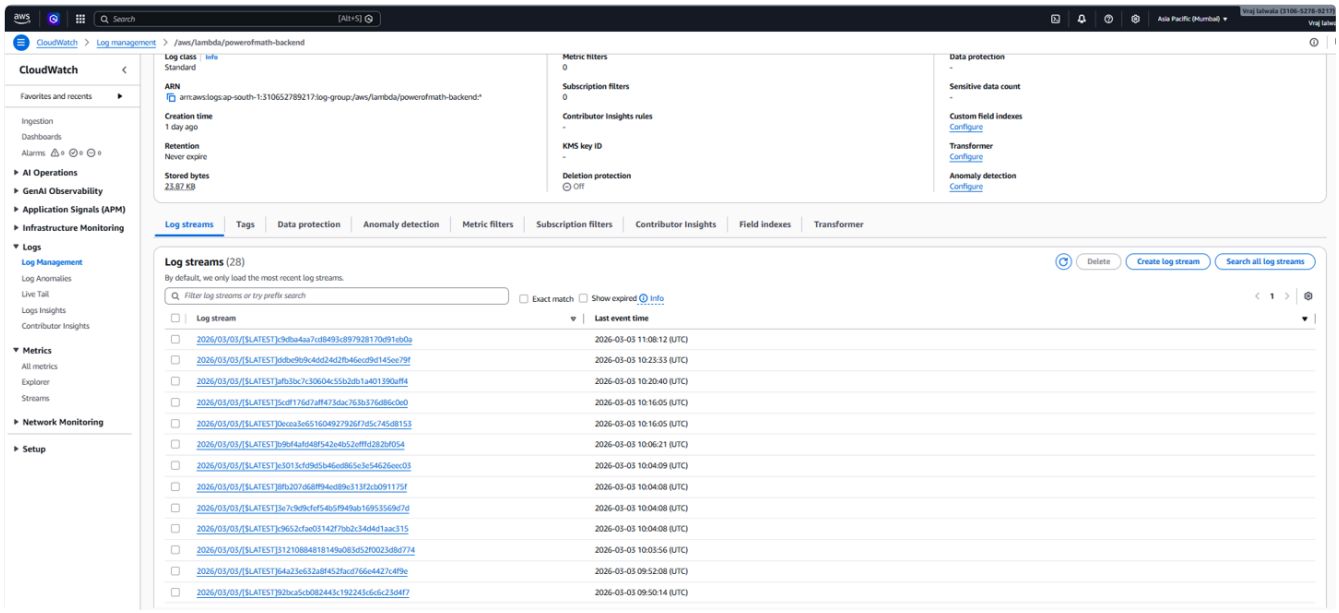






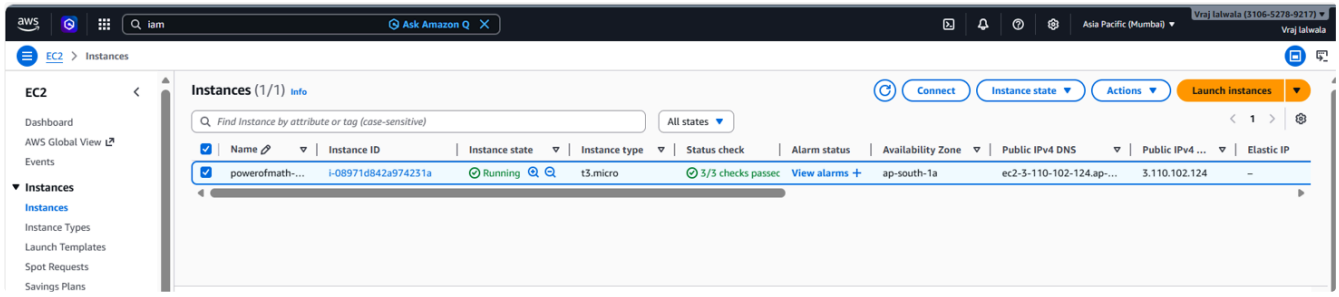
3️⃣ Reliability

- Auto-scaling in Serverless
- systemd ensures restart on crash
- DynamoDB managed service



4️⃣ Performance Efficiency

- Serverless auto scaling
- EC2 right-sized (t2.micro)
- Optimized API response time



5️⃣ Sustainability

- Serverless reduces idle compute
- Efficient resource usage

2.3 Goals and Objectives

- Deploy application using two architectures
- Implement secure HTTPS
- Integrate DynamoDB
- Apply DevOps best practices
- Demonstrate AWS architectural knowledge

2.4 Constraints and Assumptions

Constraints:

- Budget limited to AWS Free Tier
- Time constraint
- Region: ap-south-1
- Domain availability

Assumptions:

- User has AWS account
- Basic Linux knowledge
- Access to DNS configuration

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3. Architecture and Infrastructure

3.1 Infrastructure Overview

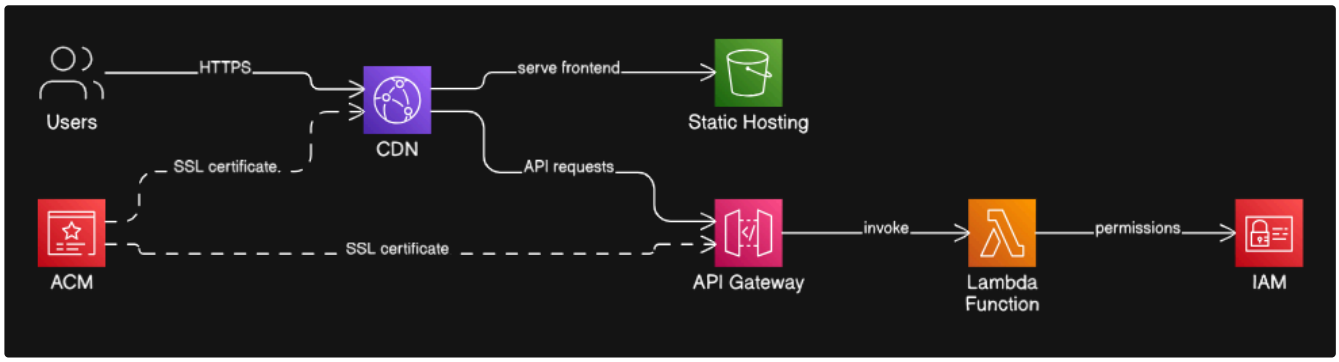
Serverless Architecture

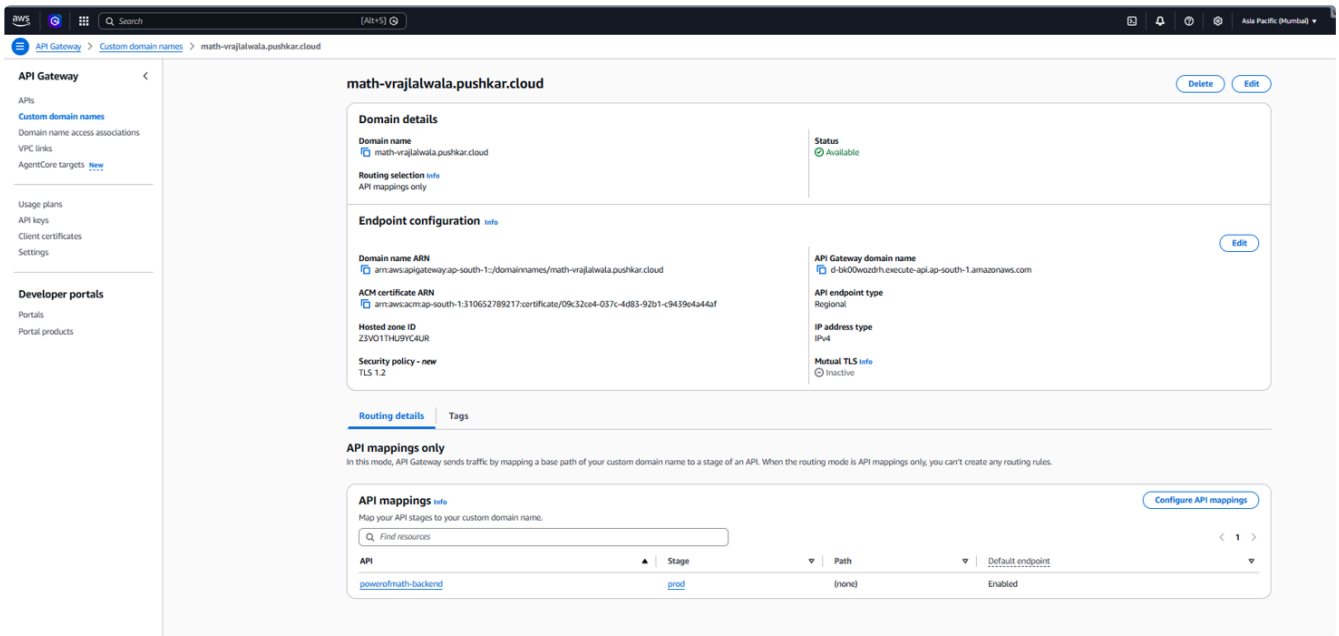
Client

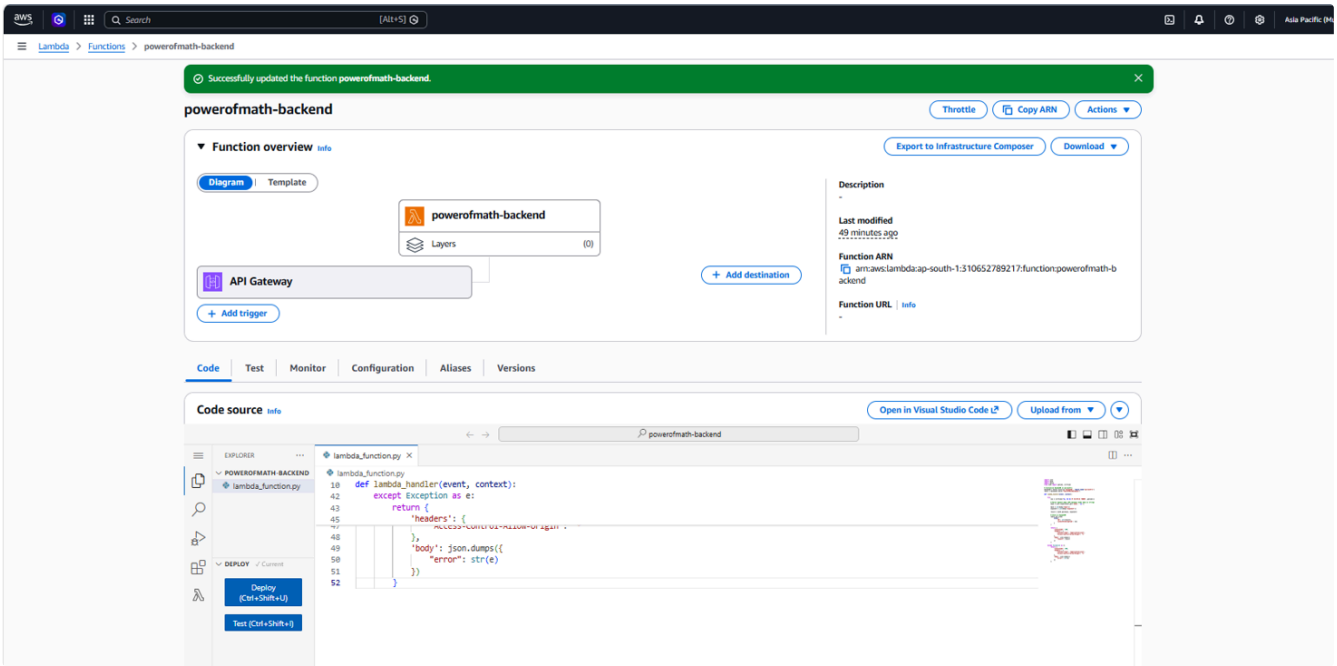
→ API Gateway

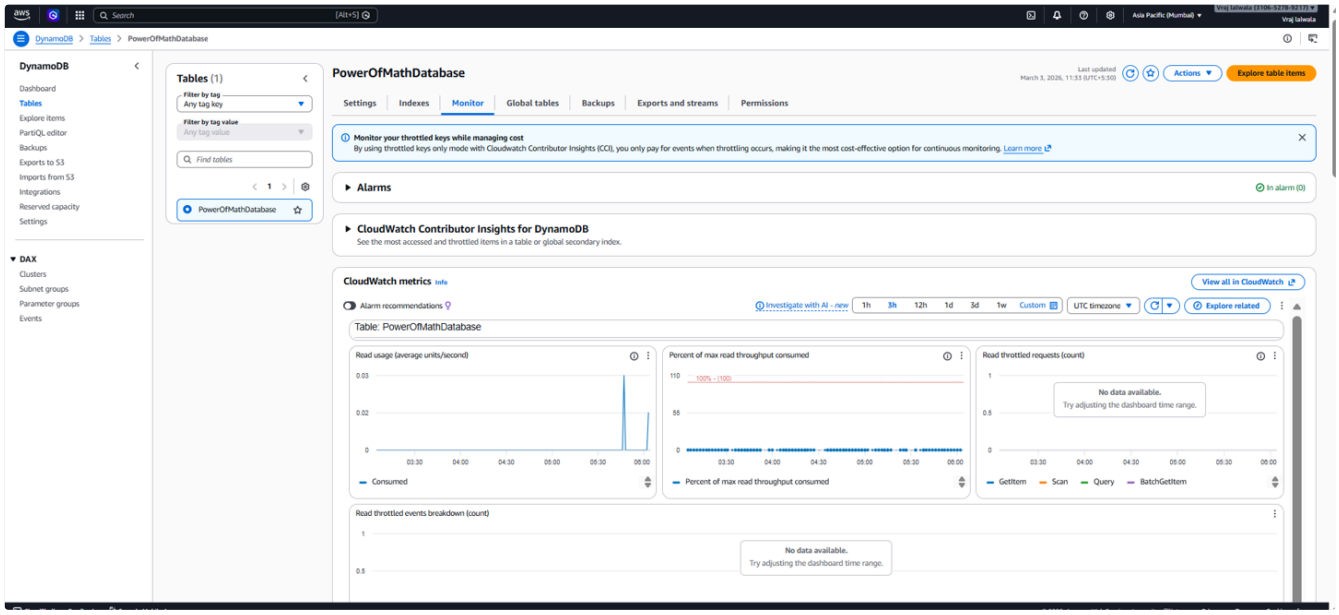
→ Lambda

→ DynamoDB









VM-Based Architecture

Client

→ Nginx (HTTPS)

→ Flask App (Port 5000)

→ DynamoDB

3.2 Environments

- Development: Local EC2 testing

- Testing: Curl-based API testing
- Production: Public HTTPS endpoints

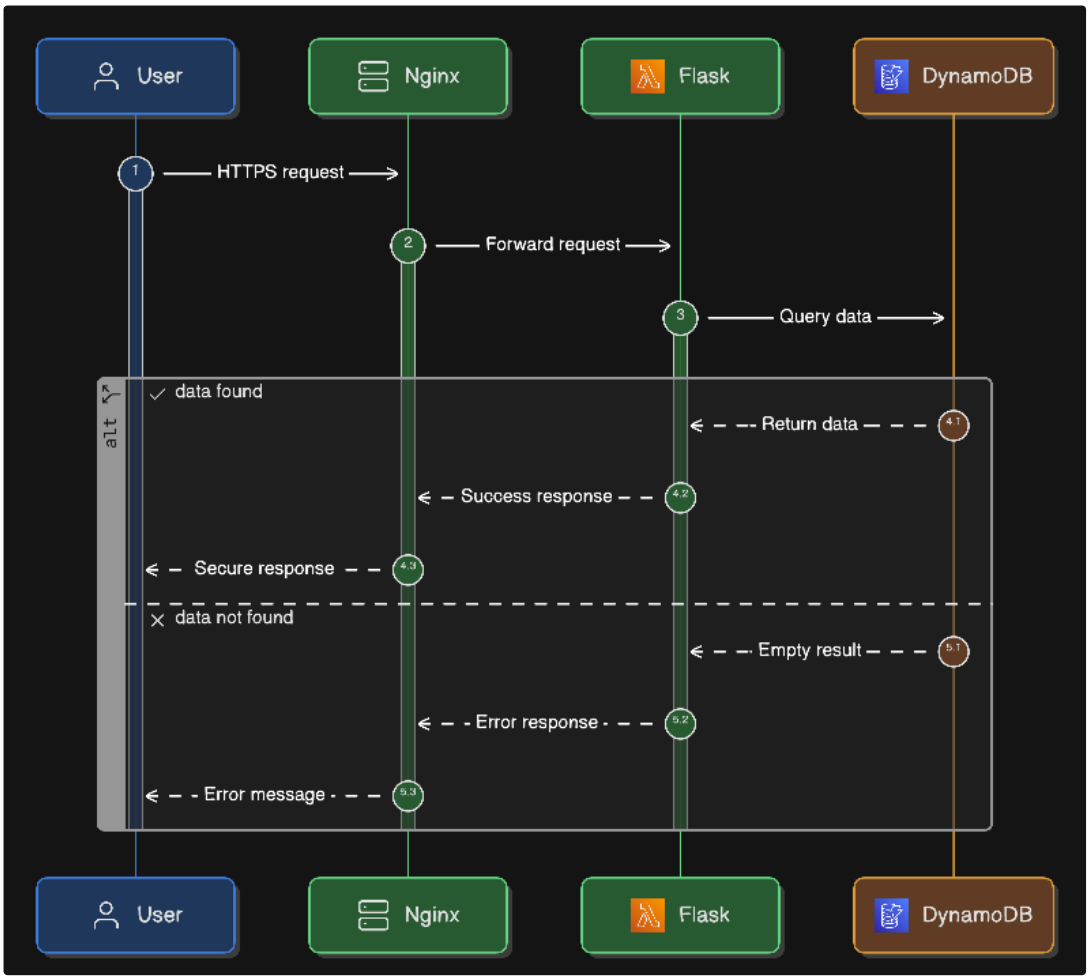
3.3 Application Architecture & User Request Flow

Serverless Flow:

User → HTTPS → API Gateway → Lambda → DynamoDB → Response

VM Flow:

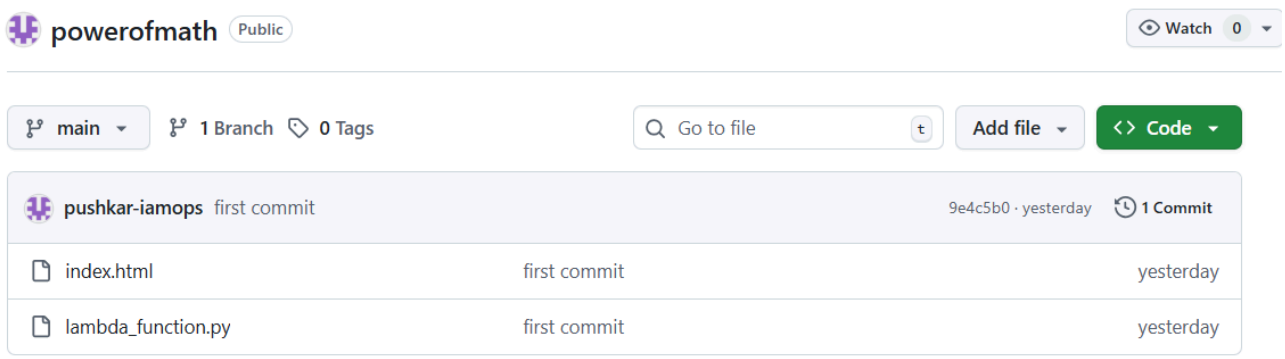
User → HTTPS → Nginx → Flask → DynamoDB → Response



3.4 Tools and Technologies

- AWS Lambda
- API Gateway
- DynamoDB
- EC2

- Nginx
- Flask
- IAM
- ACM
- Certbot
- systemd
- GitHub



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4. Development Practices

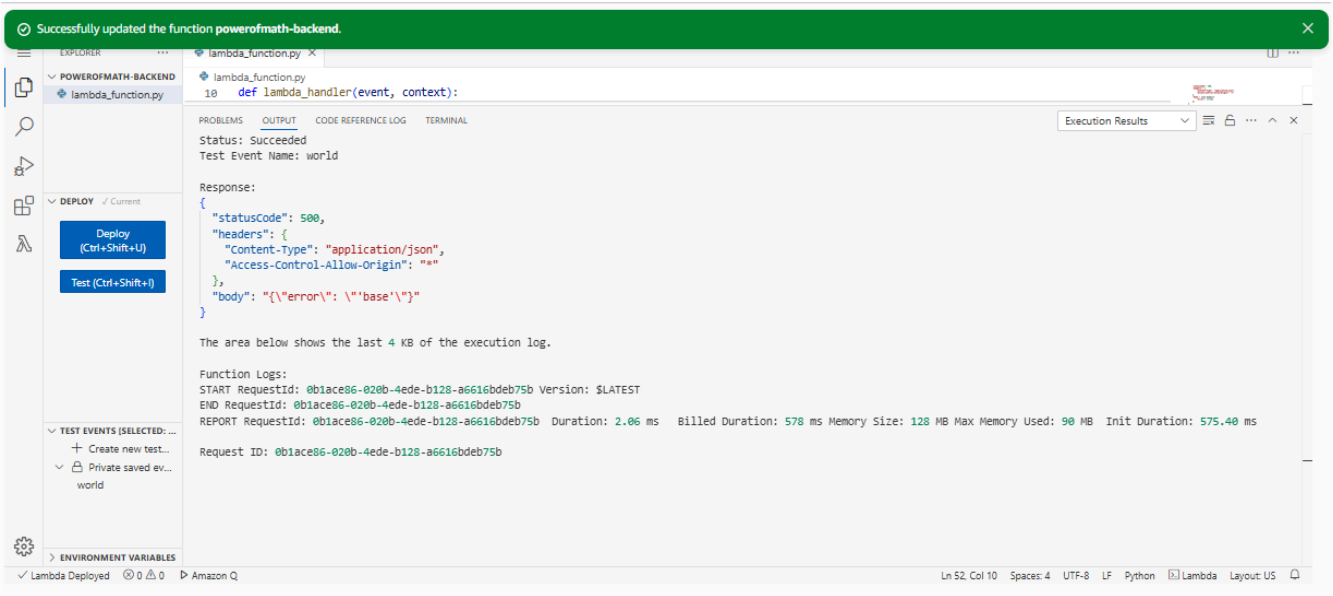
4.1 Version Control

- GitHub used
- Structured commits
- Code version maintained

4.2 Continuous Integration

Manual CI via:

- Code testing before deployment
- Lambda test events
- Curl testing



4.3 Automated Testing

- API tested via curl
- Browser testing
- DynamoDB verification

4.4 Code Quality

- Clean JSON responses
- Proper error handling
- try/except in Lambda
- Structured Flask API

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5. Deployment and Release Management

5.1 Deployment Strategy

Serverless:

- Direct deployment via Lambda console

VM:

- Manual deployment
- systemd auto-restart
- Nginx reverse proxy

5.2 Configuration Management

- systemd service file
- Nginx configuration
- IAM roles

5.3 Continuous Delivery

Deployment Flow:

Code → GitHub → AWS Console → Production Endpoint

Manual approval used before deployment.

6. Monitoring and Incident Management

6.1 Monitoring Tools

- CloudWatch Logs (Lambda)
- systemctl status (EC2)
- journalctl logs
- Nginx logs

```
(venv) ubuntu@ip-10-0-1-119:~$ journalctl -u powerofmath.service -n 50 --no-pager
Mar 02 11:55:49 ip-10-0-1-119 python[11102]: Port 5000 is in use by another program. Either identify and stop that program, or start the server with a different port.
Mar 02 11:55:49 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 02 11:55:49 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 02 11:55:50 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 5.
Mar 02 11:55:50 ip-10-0-1-119 systemd[1]: powerofmath.service: Start request repeated too quickly.
Mar 02 11:55:50 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 02 11:55:50 ip-10-0-1-119 systemd[1]: Failed to start powerofmath.service - PowerOfMath Flask App.
Mar 02 12:49:15 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 02 12:49:15 ip-10-0-1-119 python[12601]: Traceback (most recent call last):
Mar 02 12:49:15 ip-10-0-1-119 python[12601]:   File "/home/ubuntu/app.py", line 3, in <module>
Mar 02 12:49:15 ip-10-0-1-119 python[12601]:     import boto3
Mar 02 12:49:15 ip-10-0-1-119 python[12601]: ModuleNotFoundError: No module named 'boto3'
Mar 02 12:49:15 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 02 12:49:15 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 02 12:49:16 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 1.
Mar 02 12:49:16 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 02 12:49:16 ip-10-0-1-119 python[12604]: Traceback (most recent call last):
Mar 02 12:49:16 ip-10-0-1-119 python[12604]:   File "/home/ubuntu/app.py", line 3, in <module>
Mar 02 12:49:16 ip-10-0-1-119 python[12604]:     import boto3
Mar 02 12:49:16 ip-10-0-1-119 python[12604]: ModuleNotFoundError: No module named 'boto3'
Mar 02 12:49:16 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 02 12:49:16 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 02 12:49:16 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 2.
Mar 02 12:49:16 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 02 12:49:16 ip-10-0-1-119 python[12606]: Traceback (most recent call last):
Mar 02 12:49:16 ip-10-0-1-119 python[12606]:   File "/home/ubuntu/app.py", line 3, in <module>
Mar 02 12:49:16 ip-10-0-1-119 python[12606]:     import boto3
Mar 02 12:49:16 ip-10-0-1-119 python[12606]: ModuleNotFoundError: No module named 'boto3'
Mar 02 12:49:16 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 02 12:49:16 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 02 12:49:17 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 3.
Mar 02 12:49:17 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 02 12:49:17 ip-10-0-1-119 python[12609]: Traceback (most recent call last):
Mar 02 12:49:17 ip-10-0-1-119 python[12609]:   File "/home/ubuntu/app.py", line 3, in <module>
Mar 02 12:49:17 ip-10-0-1-119 python[12609]:     import boto3
Mar 02 12:49:17 ip-10-0-1-119 python[12609]: ModuleNotFoundError: No module named 'boto3'
Mar 02 12:49:17 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 02 12:49:17 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
```

6.2 Alerting

- Lambda error logs
- systemd auto restart
- Manual monitoring

6.3 Incident Management

Example incidents handled:

- Indentation errors
- IAM permission issues
- Region mismatch
- SSL misconfiguration
- CORS errors

Resolution process:

1. Identify log
2. Analyze error
3. Fix configuration
4. Restart service

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7. Security & Compliance

Implemented:

- IAM Roles
- Principle of Least Privilege
- HTTPS everywhere
- CORS configured
- Security Groups restricted
- No hardcoded credentials

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8. Cost Optimisation

Cost Inputs

- EC2 t2.micro

- Lambda requests
- DynamoDB on-demand
- API Gateway usage

Cost Model Summary

Service	Estimated Cost
Lambda	Pay per request
EC2	Free tier
DynamoDB	Free tier
API Gateway	Minimal

Business Value

- Reduced operational cost
- Auto-scaling serverless model
- Production-ready secure deployment
- Learning and skill enhancement

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9. Bonus Task – SSL Configuration & Custom Domain Implementation

9.1 Objective of Bonus Task

The objective of the bonus task was to:

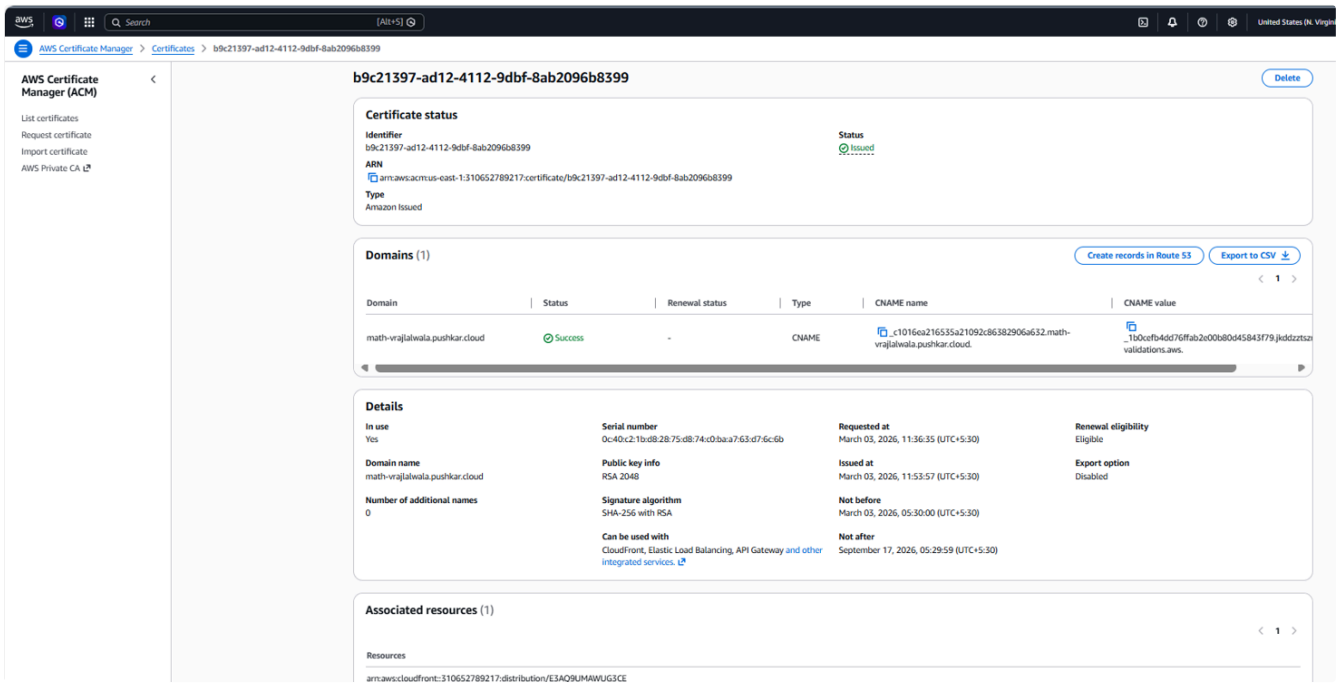
- Configure HTTPS for both deployments
- Implement custom domain integration
- Use SSL certificates properly
- Ensure secure encrypted communication

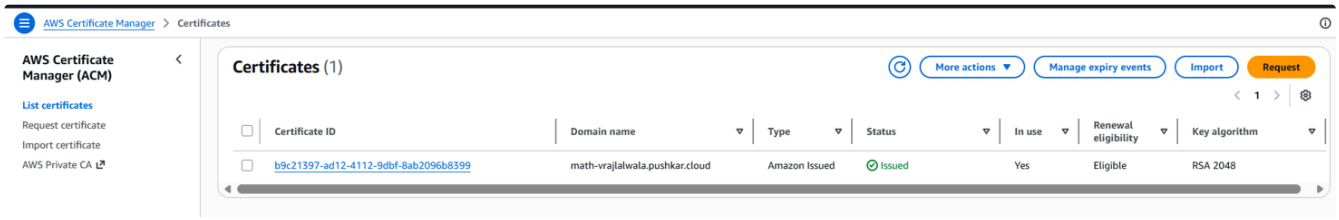
Both architectures were secured using industry-standard TLS encryption.

9.2 Serverless Architecture – SSL Configuration

9.2.1 ACM Certificate Creation

- Service: AWS Certificate Manager (ACM)
- Region: ap-south-1 (required for API Gateway Regional)
- Domain: `math-vrajlalwala.pushkar.cloud`
- Validation Method: DNS Validation

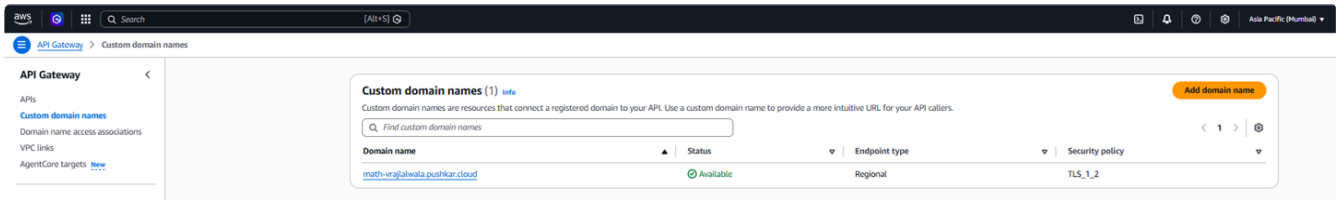




9.2.2 API Gateway Custom Domain Setup

Steps performed:

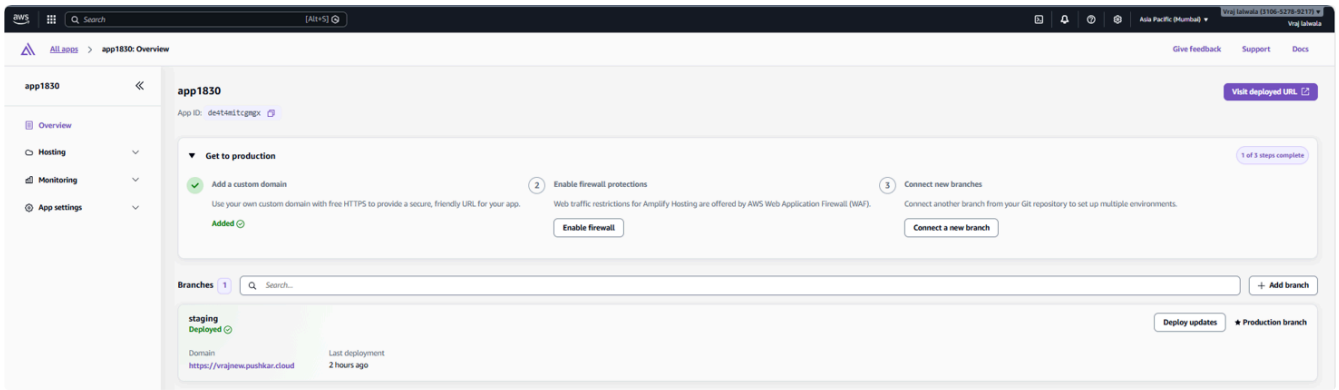
1. Created Custom Domain in API Gateway
2. Attached ACM certificate
3. Selected Regional Endpoint
4. Mapped stage (prod) to domain



9.2.3 DNS Configuration

- Added CNAME record:

```
Name: math-vrajlalwala.pushkar.cloud
Value: d-xxxx.execute-api.ap-south-1.amazonaws.com
```

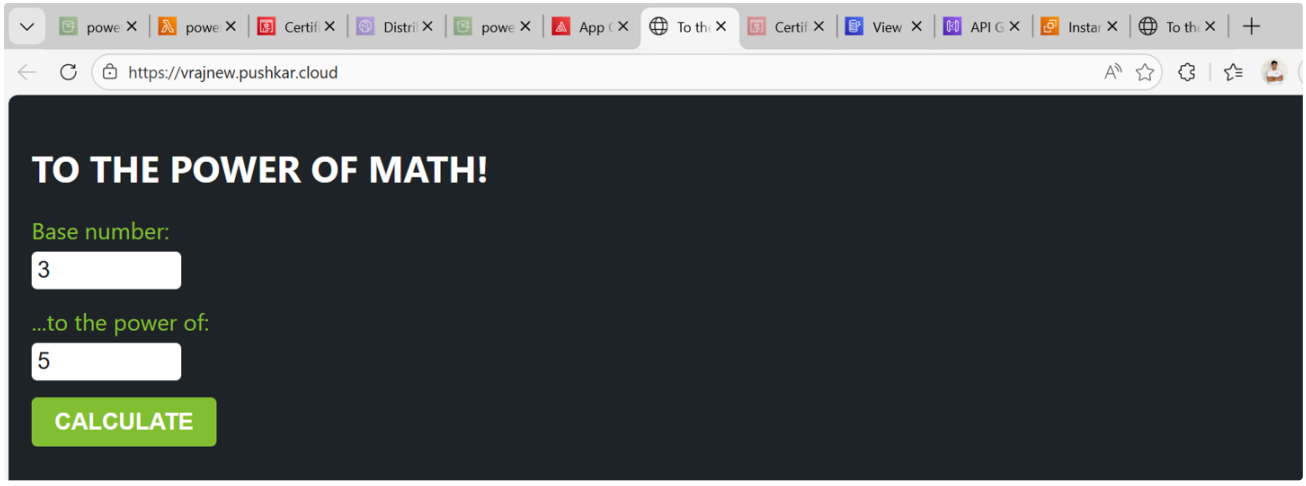


9.2.4 HTTPS Validation

Verified:

- Secure lock icon in browser
- No SSL warning
- API accessible via HTTPS

📸 Add:

- 

```
PS C:\Users\vraj.lalwala> nslookup vrajnew.pushkar.cloud
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    d3kr9onigarw0j.cloudfront.net
Addresses:  2600:9000:2652:f000:10:4c51:34c0:93a1
          2600:9000:2652:2400:10:4c51:34c0:93a1
          2600:9000:2652:5400:10:4c51:34c0:93a1
          2600:9000:2652:f800:10:4c51:34c0:93a1
          2600:9000:2652:9800:10:4c51:34c0:93a1
          2600:9000:2652:c00:10:4c51:34c0:93a1
          2600:9000:2652:6a00:10:4c51:34c0:93a1
          2600:9000:2652:f400:10:4c51:34c0:93a1
          18.239.111.116
          18.239.111.58
          18.239.111.123
          18.239.111.126
Aliases:  vrajnew.pushkar.cloud

PS C:\Users\vraj.lalwala>
```

9.3 VM-Based Architecture – SSL Configuration

9.3.1 Domain Setup

Configured subdomain:

```
api-vrajlalwala.pushkar.cloud
```

DNS A record pointing to EC2 public IP.

```
PS C:\Users\vraj.lalwala> nslookup api-vrajlalwala.pushkar.cloud
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    api-vrajlalwala.pushkar.cloud
Address:  3.110.102.124

PS C:\Users\vraj.lalwala>
```

9.3.2 Nginx Reverse Proxy Configuration

Configured:

```
server {
    server_name api-vrajlalwala.pushkar.cloud;

    location /calculate {
        proxy_pass http://127.0.0.1:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

9.3.3 Let's Encrypt SSL Configuration

Used Certbot:

```
sudo certbot --nginx
```

Certificate automatically installed and Nginx updated.

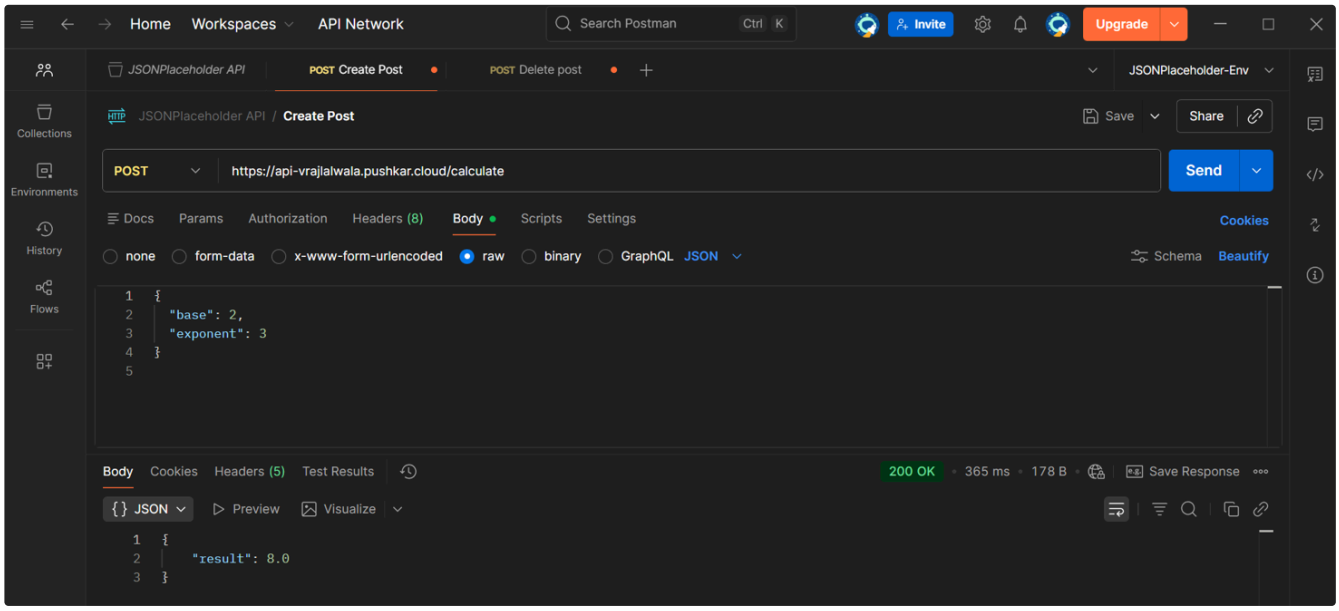
```
(venv) ubuntu@ip-10-0-1-119:~$ sudo certbot --nginx
Saving debug log to /var/log/letsencrypt/letsencrypt.log

Which names would you like to activate HTTPS for?
We recommend selecting either all domains, or all domains in a VirtualHost/server block.
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
1: api-vrajlalwala.pushkar.cloud
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Select the appropriate numbers separated by commas and/or spaces, or leave input
blank to select all options shown (Enter 'c' to cancel):
Certificate not yet due for renewal

You have an existing certificate that has exactly the same domains or certificate name you requested and isn't close to expiry.
(ref: /etc/letsencrypt/renewal/api-vrajlalwala.pushkar.cloud.conf)

What would you like to do?
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
1: Attempt to reinstall this existing certificate
2: Renew & replace the certificate (may be subject to CA rate limits)
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Select the appropriate number [1-2] then [enter] (press 'c' to cancel): 2
Renewing an existing certificate for api-vrajlalwala.pushkar.cloud

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/api-vrajlalwala.pushkar.cloud/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/api-vrajlalwala.pushkar.cloud/privkey.pem
This certificate expires on 2026-06-01.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for api-vrajlalwala.pushkar.cloud to /etc/nginx/sites-enabled/default
Your existing certificate has been successfully renewed, and the new certificate has been installed.

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
(venv) ubuntu@ip-10-0-1-119:~$
```

9.3.4 HTTPS Validation (VM)

Validated:

- Secure HTTPS endpoint

- Proper reverse proxy
- Flask app working over SSL




```
(venv) ubuntu@ip-10-0-1-119:~$ curl -X POST https://api-vrajlalwala.pushkar.cloud/calculate \
-H "Content-Type: application/json" \
-d '{"base":3,"exponent":5}'
{"result":243.0}
```

9.4 Security Benefits Achieved

✔ Data Encryption in Transit

All communication encrypted using TLS.

✔ Secure Identity Verification

Certificates verified by trusted CA.

✔ No Plain HTTP

Redirected HTTP to HTTPS.

✔ Compliance Ready

Aligned with security best practices.

9.5 Comparison of SSL Implementation

Feature	Serverless	VM-Based
SSL Tool	ACM	Certbot
Certificate Management	Fully Managed	Manual Renewal
Renewal	Automatic	Automated via Certbot
Integration	API Gateway	Nginx
Effort	Low	Medium

9.6 Challenges Faced During Bonus Task

- SSL mismatch error
- Wrong region ACM certificate
- DNS propagation delay
- Nginx 502 error
- Port conflicts
- JSON response mismatch

Resolution Approach:

- Verified region alignment
- Used nslookup for debugging
- Restarted services
- Verified systemd logs
- Tested via curl

```
Mar 03 09:46:20 ip-10-0-1-119 python[29325]: 27.116.48.171 - - [03/Mar/2026 09:46:20] "GET / HTTP/1.1" 200 -
Mar 03 09:55:00 ip-10-0-1-119 systemd[1]: Stopping powerofmath.service - PowerOfMath Flask App...
Mar 03 09:55:00 ip-10-0-1-119 systemd[1]: powerofmath.service: Deactivated successfully.
Mar 03 09:55:00 ip-10-0-1-119 systemd[1]: Stopped powerofmath.service - PowerOfMath Flask App.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 03 09:55:01 ip-10-0-1-119 python[29374]:   File "/home/ubuntu/app.py", line 33
Mar 03 09:55:01 ip-10-0-1-119 python[29374]:     return jsonify({
Mar 03 09:55:01 ip-10-0-1-119 python[29374]:                    ^
Mar 03 09:55:01 ip-10-0-1-119 python[29374]: IndentationError: unindent does not match any outer indentation level
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 1.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 03 09:55:01 ip-10-0-1-119 python[29377]:   File "/home/ubuntu/app.py", line 33
Mar 03 09:55:01 ip-10-0-1-119 python[29377]:     return jsonify({
Mar 03 09:55:01 ip-10-0-1-119 python[29377]:                    ^
Mar 03 09:55:01 ip-10-0-1-119 python[29377]: IndentationError: unindent does not match any outer indentation level
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 2.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 03 09:55:01 ip-10-0-1-119 python[29379]:   File "/home/ubuntu/app.py", line 33
Mar 03 09:55:01 ip-10-0-1-119 python[29379]:     return jsonify({
Mar 03 09:55:01 ip-10-0-1-119 python[29379]:                    ^
Mar 03 09:55:01 ip-10-0-1-119 python[29379]: IndentationError: unindent does not match any outer indentation level
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 3.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 03 09:55:01 ip-10-0-1-119 python[29384]:   File "/home/ubuntu/app.py", line 33
Mar 03 09:55:01 ip-10-0-1-119 python[29384]:     return jsonify({
Mar 03 09:55:01 ip-10-0-1-119 python[29384]:                    ^
Mar 03 09:55:01 ip-10-0-1-119 python[29384]: IndentationError: unindent does not match any outer indentation level
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 4.
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: Started powerofmath.service - PowerOfMath Flask App.
Mar 03 09:55:01 ip-10-0-1-119 python[29386]:   File "/home/ubuntu/app.py", line 33
Mar 03 09:55:01 ip-10-0-1-119 python[29386]:     return jsonify({
Mar 03 09:55:01 ip-10-0-1-119 python[29386]:                    ^
Mar 03 09:55:01 ip-10-0-1-119 python[29386]: IndentationError: unindent does not match any outer indentation level
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Main process exited, code=exited, status=1/FAILURE
Mar 03 09:55:01 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 03 09:55:02 ip-10-0-1-119 systemd[1]: powerofmath.service: Scheduled restart job, restart counter is at 5.
Mar 03 09:55:02 ip-10-0-1-119 systemd[1]: powerofmath.service: Start request repeated too quickly.
Mar 03 09:55:02 ip-10-0-1-119 systemd[1]: powerofmath.service: Failed with result 'exit-code'.
Mar 03 09:55:02 ip-10-0-1-119 systemd[1]: Failed to start powerofmath.service - PowerOfMath Flask App.
(venv) ubuntu@ip-10-0-1-119:~$ nano app.py
```

9.7 Final Secure Endpoints

Serverless:

```
https://math-vrajlalwala.pushkar.cloud/calculate
```

VM-Based:

```
https://api-vrajlalwala.pushkar.cloud/calculate
```

Both endpoints successfully return JSON responses over HTTPS.

---

10. Implementation Timeline

Phase 1 – Serverless Setup

- DynamoDB creation

- Lambda deployment
- API Gateway configuration
- SSL configuration

Phase 2 – VM Deployment

- EC2 launch
- Flask setup
- systemd configuration
- Nginx + SSL setup

Final Delivery

Completed successfully with both architectures operational.

---

11. Continuous Improvement

11.1 Feedback & Retrospectives

Issues identified:

- JSON mismatch
- Indentation errors
- Region configuration
- SSL mismatch

Lessons learned:

- Always validate region
- Always test via curl
- Always check logs

11.2 Iterative Enhancements

Future improvements:

- Add CI/CD pipeline
- Add Load Balancer
- Add Auto Scaling
- Add Monitoring Dashboard

12. Conclusion

12.1 Summary

The PowerOfMath application was successfully deployed using:

- Serverless AWS architecture
- Traditional VM-based architecture

Both deployments:

- Are secure
- Use HTTPS
- Use DynamoDB
- Follow DevOps best practices
- Align with AWS Well-Architected Framework

12.2 Next Steps

- Implement CI/CD using GitHub Actions
- Add Auto Scaling Group
- Add CloudFront
- Add centralized monitoring

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13. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____